

Functional Test Cases: State Transition Testing

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed from a black-box, user-centric perspective based on SRS requirements for entities with distinct states. Actual results and status are to be filled during test execution. Refer to the SRS for complete requirements.

I.E Functional Test Cases: State Transition Testing

Table 1: State Transition Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Entity: Borrowal Record (Based on 'status' attribute)			
TC_STATE_BORROW_001	Valid Transition: (None) to 'Borrowed'	- Login as a Member. - Navigate to the Borrowal Management page. - Create a new borrowal record for an available book. - Submit the form.	- A new borrowal record is successfully created. - The status of the new record is set to "Borrowed". - A success message is displayed.
TC_STATE_BORROW_002	Valid Transition: 'Borrowed' to 'Returned'	- Precondition: A book is in 'Borrowed' status by a member. - Login as Librarian. - Navigate to Borrowal Management. - Locate the 'Borrowed' record. - Update the record's status to 'Returned' before its due date.	- The system accepts the update. - The borrowal record's status changes from "Borrowed" to "Returned". - A success message is displayed.
TC_STATE_BORROW_003	Valid Transition: 'Borrowed' to 'Overdue'	- Precondition: A book is in 'Borrowed' status by a member. - Manually adjust system time or the record's due date to be in the past. - As a Librarian, view the borrowal record. (Or trigger a system check for overdue items).	- The borrowal record's status automatically (or upon view) changes from "Borrowed" to "Overdue". - The status is displayed correctly in the UI.
TC_STATE_BORROW_004	Valid Transition: 'Overdue' to 'Returned'	- Precondition: A book is in 'Overdue' status. - Login as Librarian. - Locate the 'Overdue' borrowal record. - Update the record's status to 'Returned'.	- The system accepts the update. - The borrowal record's status changes from "Overdue" to "Returned". - A success message is displayed.
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Test Case ID	Test Title	Test Steps	Expected Results
TC_STATE_BORROW_005	Invalid Transition: 'Returned' to 'Borrowed'	• Precondition: A borrowal record has a 'Returned' status. • As a Librarian, attempt to edit the same record and change its status back to 'Borrowed'.	• The system should reject the change. • The UI should not provide an option for this transition, or it should result in an error if attempted via API. • An appropriate error message is displayed (e.g., "Cannot revert a returned item").
Entity: Book (Based on 'isAvailable' attribute)			
TC_STATE_BOOK_001	Valid Transition: 'Available' to 'Unavailable'	• Precondition: A book exists with 'isAvailable' status set to true. • As a Member, create a new borrowal for that specific book. • After the borrowal is confirmed, view the details of that book.	• The borrowal transaction is successful. • The 'isAvailable' status of the book is updated to false. • The book's listing correctly shows it as unavailable.
TC_STATE_BOOK_002	Valid Transition: 'Unavailable' to 'Available'	• Precondition: A book has 'isAvailable' status set to false due to being borrowed. • As a Librarian, locate the corresponding borrowal record. • Update the borrowal record status to 'Returned'. • View the details of the book again.	• The 'isAvailable' status of the book is updated to true. • The book's listing correctly shows it as available for borrowing.
TC_STATE_BOOK_003	Invalid Transition: Attempt to borrow an 'Unavailable' book	• Precondition: A book has 'isAvailable' status set to false. • As a Member, attempt to create a new borrowal for that specific book.	• The system prevents the borrowal request from being submitted. • An appropriate error message is displayed (e.g., "Book is currently unavailable"). • No new borrowal record is created.
Entity: User Session (Based on login/logout actions)			
TC_STATE_SESSION_001	Valid Transition: 'Logged-Out' to 'Logged-In'	• Start from a logged-out state (e.g., in an incognito window). • Navigate to the login page. • Enter valid credentials for a 'Member'. • Click 'Login'.	• User session is established. • User is redirected to the member landing page (e.g., /books). • User can access member-specific pages.
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Test Case ID	Test Title	Test Steps	Expected Results
TC_STATE_SESSION_002	Valid Transition: 'Logged-In' to 'Logged-Out'	• Precondition: A user is logged in. • Click the 'Logout' button/link.	• The user's session is terminated. • User is redirected to the login page.
TC_STATE_SESSION_003	Invalid Transition: Accessing protected page when 'Logged-Out'	• Start from a logged-out state. • Attempt to directly navigate to a protected URL (e.g., /borrowals, /users).	• Access to the protected page is denied. • User is redirected to the login page.
TC_STATE_SESSION_004	State Persistence: Verify session remains 'Logged-In' after page refresh	• Login as any user. • Navigate to any page other than the login page. • Refresh the web browser page.	• The user remains logged in. • The user stays on the same page. • No redirection to the login page occurs.